

AUDIOSHIELD FORENSICS

Audio Forensic Analysis Report

Report ID	ASF-20260914-0003
Generated	2026-09-14 09:55:15 India Standard Time
File analyzed	LA_T_1000137.flac

This report presents separate evidence streams and conservative forensic indicators. It is not a substitute for expert examination of the original evidence.

1. Executive Summary

Overall risk assessment: **medium**
Overall assessment: **inconclusive** | Confidence: **0.0**

Key findings

- AI consensus: unknown (risk: medium).
 - Metadata risk: medium; 3 finding(s).
 - Signal risk: low; 0 anomaly/anomalies.
 - AI evidence unavailable: AI results were not provided.
- Recommendation: Evidence is mixed or incomplete; requires further investigation.

2. File Information

Property	Value
File name	LA_T_1000137.flac
File size (bytes)	48751
Format	flac
Duration (seconds)	2.500375
Sample rate (Hz)	16000
Channels	1
Codec	flac
Bitrate (bits/sec)	155979
Creation time	2018:12:11 23:45:52+05:30
Modification time	2026:08:30 18:07:06+05:30

3. Metadata Forensics

Extracted metadata

Field	Value
File Name	LA_T_1000137.flac
File Size	48751
Format	flac
Duration	2.500375
Codec	flac
Sample Rate	16000
Channels	1
Bitrate	155979
Encoder	Not available

Field	Value
Creation Time	2018:12:11 23:45:52+05:30
Modification Time	2026:08:30 18:07:06+05:30
Device Info	Not available
Software	Not available
Comments	Not available

Risk score: **medium**

Findings

Severity	Field	Description
medium	timestamps	Creation and modification timestamps are more than 30 days apart; this is inconsistent.
low	device	Device information is missing; provenance cannot be verified from metadata alone.
low	encoder	Encoder or software information is missing.

Metadata indicators are not conclusive on their own and require further investigation.

4. Signal Forensics

Spectral features

Feature	Mean	Std	Min	Max
Spectral Centroid	887.0816367725992	1054.6279422380699	0.0	4697.450802419621
Spectral Bandwidth	877.0676937762518	900.0006089737436	0.0	2613.788826875332
Spectral Rolloff	1841.5107484076434	2013.5102089197565	0.0	6625.0
Spectral Flatness	0.5007465512804675	0.4904759870285128	0.0007324606413021684	1.0000007152557373
Spectral Contrast	18.575692857382748	6.0476026533270915	3.8230853257893145	33.0153944278325

Temporal features

Feature	Mean	Std	Min	Max
Zero Crossing Rate	0.05530329418789809	0.0919807994839511	0.0	0.3544921875
Rms Energy	0.04569135930751724	0.06330303652752556	0.0	0.21424275636672974
Pitch	125.7816213227827	27.35128909669236	65.40639132514966	192.63059450929475

Risk score: **low**

Anomalies

No signal-level anomalies were reported.

5. Evidence Fusion

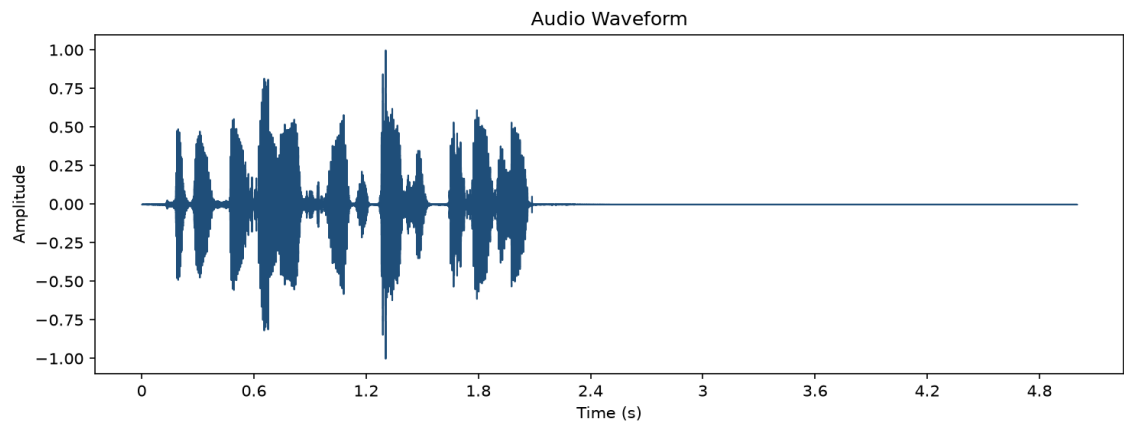
Evidence source	Risk / result
AI	unknown (risk: medium)
Metadata	medium
Signal	low
Overall	medium

Assessment: **inconclusive** | Confidence: **0.0**

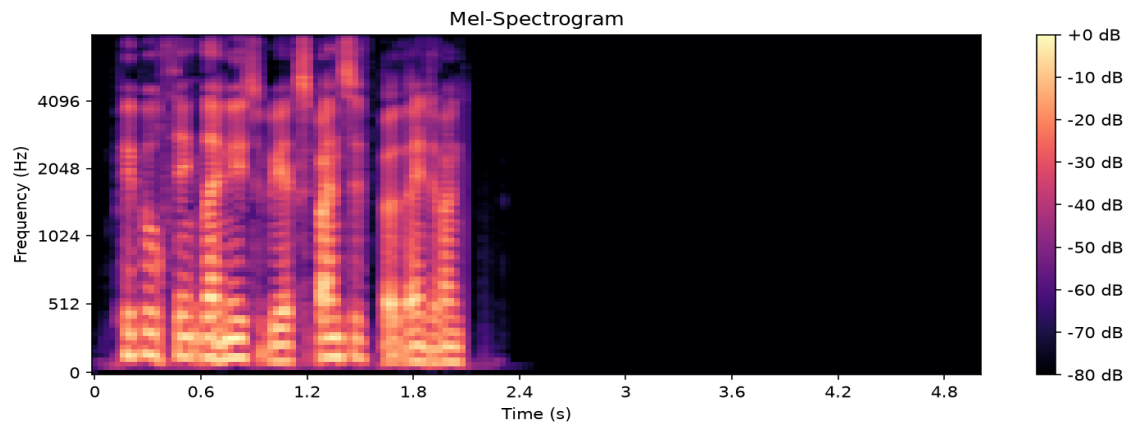
Reasoning: Source risks are AI=medium, metadata=medium, signal=low; applied documented rule-based logic.

Recommendation: Evidence is mixed or incomplete; requires further investigation.

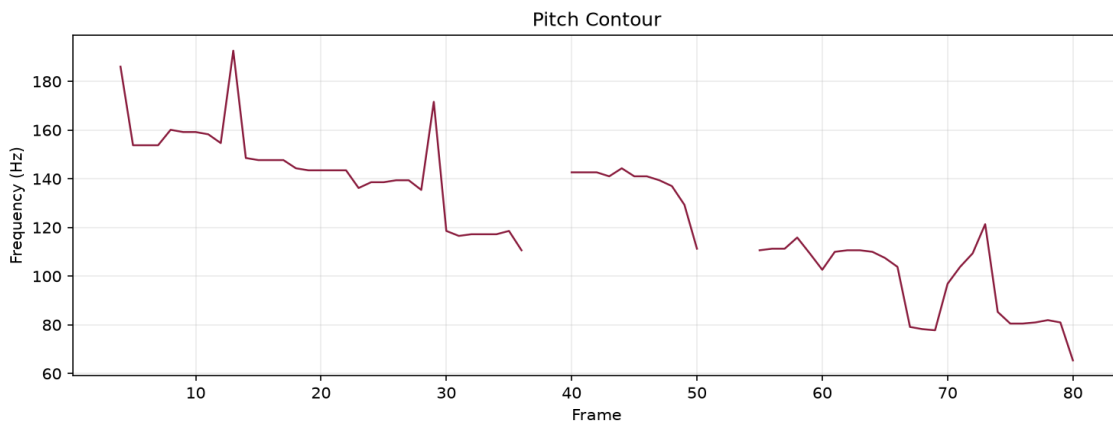
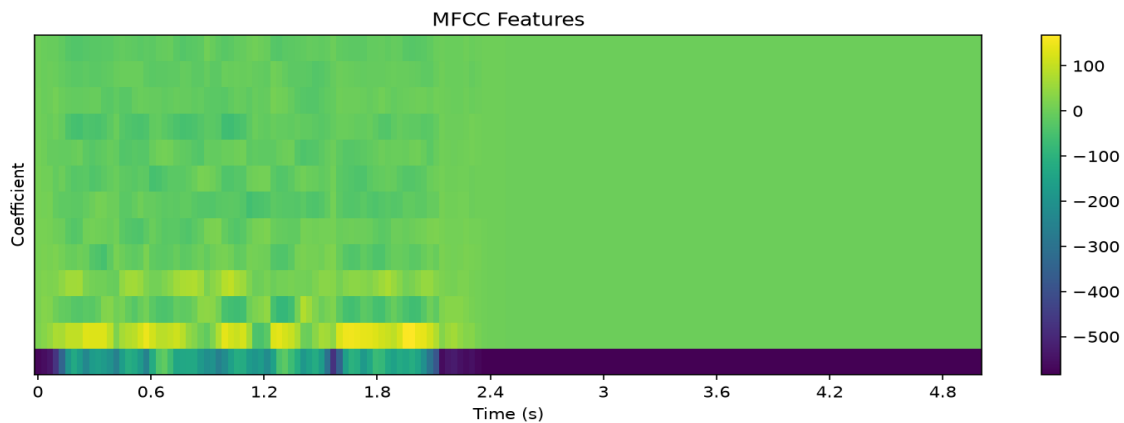
6. Visual Analysis



Waveform: Time-domain waveform.



Spectrogram: Frequency content over time.



7. Conclusion

The combined evidence indicates an overall assessment of **inconclusive** with a reported risk level of **medium**. These are indicators from the available reports and should be interpreted with the original audio and chain of custody.

Limitations: AI results may be unavailable; metadata can be edited or lost; signal features are influenced by recording conditions and preprocessing; and this report does not establish authenticity by itself.

8. Appendix

Raw metadata report

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{ "metadata": { "file_name": "LA_T_1000137.flac", "file_size": "48751", "format": "flac", "duration": "2.500375", "codec": "flac", "sample_rate": "16000", "channels": 1, "bitrate": "155979", "encoder": null, "creation_time": "2018:12:11 23:45:52+05:30", "modification_time": "2026:08:30 18:07:06+05:30", "device_info": null, "software": null, "comments": null, "raw_ffprobe": { "streams": [ { "index": 0, "codec_name": "flac", "codec_long_name": "FLAC (Free Lossless Audio Codec)", "codec_type": "audio", "codec_tag_string": "[0][0][0][0]", "codec_tag": "0x0000", "mime_codec_string": "flac", "sample_fmt": "s16", "sample_rate": "16000", "channels": 1, "channel_layout": "mono", "bits_per_sample": 0, "initial_padding": 0, "r_frame_rate": "0/0", "avg_frame_rate": "0/0", "time_base": "1/16000", "start_pts": 0, "start_time": "0.000000", "duration_ts": 40006, "duration": "2.500375", "bits_per_raw_sample": "16", "extradata_size": 34, "disposition": { "default": 0, "dub": 0, "original": 0, "comment": 0, "lyrics": 0, "karaoke": 0, "forced": 0, "hearing_impaired": 0, "visual_impaired": 0, "clean_effects": 0, "attached_pic": 0, "timed_thumbnails": 0, "non_diegetic": 0, "captions": 0, "descriptions": 0, "metadata": 0, "dependent": 0, "still_image": 0, "multilayer": 0 } } ], "format": { "filename": "data\\LA\\ASVspoof2019_LA_train\\flac\\LA_T_1000137.flac", "nb_streams": 1, "nb_programs": 0, "nb_stream_groups": 0, "format_name": "flac", "format_long_name": "raw FLAC", "start_time": "0.000000", "duration": "2.500375", "size": "48751", "bit_rate": "155979", "probe_score": 100 } }, "raw_exiftool": { "SourceFile": "data/LA/ASVspoof2019_LA_train/flac/LA_T_1000137.flac", "ExifToolVersion": 13.59, "FileName": "LA_T_1000137.flac", "Directory": "data/LA/ASVspoof2019_LA_train/flac", "FileSize": "49 kB", "ZoneIdentifier": "Exists", "FileModifyDate": "2026:08:30 18:07:06+05:30", "FileAccessDate": "2026:09:13 19:49:45+05:30", "FileCreateDate": "2018:12:11 23:45:52+05:30", "FilePermissions": "-rw-rw-rw-", "FileType": "FLAC", "FileTypeExtension": "flac", "MIMETYPE": "audio/flac", "BlockSizeMin": 1152, "BlockSizeMax": 1152, "FrameSizeMin": 226, "FrameSizeMax": 1904, "SampleRate": 16000, "Channels": 1, "BitsPerSample": 16,
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"TotalSamples": 40006, "MD5Signature": "2fe8bb74b749bdb6467bfab7723cc8b8", "Vendor": "reference libFLAC 1.3.0
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"155979", "encoder": null, "creation_time": "2018:12:11 23:45:52+05:30", "modification_time": "2026:08:30
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alone." }, { "field": "encoder", "has_issue": true, "severity": "low", "description": "Encoder or software
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"missing_fields": [ "device_info", "encoder", "software", "comments" ] }, "report": "Metadata Forensics
Report\nRisk score: medium\nSuspicious fields: timestamps\nMissing fields: device_info, encoder, software,
comments\n\nFindings:\n- [medium] timestamps: Creation and modification timestamps are more than 30 days
apart; this is inconsistent.\n- [low] device: Device information is missing; provenance cannot be verified
from metadata alone.\n- [low] encoder: Encoder or software information is missing.\nMetadata findings are
indicators only and require further investigation." }

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Raw signal report

```

{ "spectral_features": { "spectral_centroid": { "mean": 887.0816367725992, "std": 1054.6279422380699, "min":
0.0, "max": 4697.450802419621 }, "spectral_bandwidth": { "mean": 877.0676937762518, "std": 900.0006089737436,
"min": 0.0, "max": 2613.788826875332 }, "spectral_rolloff": { "mean": 1841.5107484076434, "std":
2013.5102089197565, "min": 0.0, "max": 6625.0 }, "spectral_flatness": { "mean": 0.5007465512804675, "std":
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18.575692857382748, "std": 6.0476026533270915, "min": 3.8230853257893145, "max": 33.0153944278325 },
"temporal_features": { "zero_crossing_rate": { "mean": 0.05530329418789809, "std": 0.0919807994839511, "min":
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65.40639132514966, "max": 192.63059450929475 } }, "anomalies": { "anomalies": [], "risk_score": "low",
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} }, "plots": [ "results\\signal_plots\\waveform.png", "results\\signal_plots\\spectrogram.png",
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"summary": "Signal risk is low. No broad signal-level anomalies identified." }

```

Methodology

This report compiles the supplied AI, metadata, signal, and fusion JSON outputs. No new model inference is performed during report generation. Evidence sources remain separated to preserve interpretability.